

Supplementary Material on Evolutionary Payoff Analysis for Data Validation and Proxy Nodes

I. EVOLUTIONARY ANALYSIS OF PAYOFFS FOR DATA, VALIDATION, AND PROXY NODES UNDER DIFFERENT STRATEGIES

Referencing the payoff matrix in Table I, the payoff of data nodes for choosing honest behavior U_{11} , the payoff for choosing malicious behavior U_{12} , and the average expected payoff for data nodes choosing between the two strategies $\bar{U}_1$ are calculated as follows:

$$\begin{aligned} U_{11} &= yz(\phi R_D - C_D) + (1-y)z(\phi R_D - C_D) + y(1-z)(\phi R_D - C_D + r) + (1-y)(1-z)(\phi R_D - C_D) \\ &= \phi R_D - C_D + yr - yzr, \end{aligned} \quad (1)$$

$$\begin{aligned} U_{12} &= yz(-C_D - P_D) + (1-y)z(-C_D - C_C - P_D) + y(1-z)(\phi R_D - C_D) + (1-y)(1-z)(R_{MD} - C_D - C_C) \\ &= y(C_C - R_{MD} + \phi R_D + zR_{MD} - z\phi R_D) + R_{MD} - C_D - C_C - zP_D - zR_{MD}, \end{aligned}$$

and

$$\bar{U}_1 = xU_{11} + (1-x)U_{12}, \quad (3)$$

The payoff of validator nodes for choosing honest behavior U_{21} , the payoff for choosing malicious behavior U_{22} , and the average expected payoff for validator nodes choosing between the two strategies $\bar{U}_2$ are calculated as follows:

$$\begin{aligned} U_{21} &= xz(R_V - C_V) + (1-x)z(R_V - C_V) + x(1-z)(R_V - C_V + r) + (1-x)(1-z)(R_V - C_V) \\ &= R_V - C_V + xr - xzr, \end{aligned} \quad (4)$$

$$\begin{aligned} U_{22} &= xz(-C_V - P_V) + (1-x)z(-C_V - P_V) + x(1-z)(R_V - C_V) + (1-x)(1-z)(R_{MV} - C_V) \\ &= (R_V - R_{MV} + zR_{MV} - zR_V)x + R_{MV} - C_V - zP_V - zR_{MV}, \end{aligned} \quad (5)$$

and

$$\bar{U}_2 = yU_{21} + (1-y)U_{22}. \quad (6)$$

Similarly, the payoff of proxy nodes for choosing honest behavior U_{31} , the payoff for choosing malicious behavior U_{32} , and the average expected payoff for proxy nodes choosing between the two strategies $\bar{U}_3$ are calculated as follows:

$$\begin{aligned} U_{31} &= xy(R_B - C_B - C_S) + (1-x)y(R_B - C_B - C_S + P_D) + x(1-y)(R_B - C_B - C_S + P_V) \\ &\quad + (1-x)(1-y)(R_B - C_B - C_S + P_D + P_V) \\ &= R_B - C_B - C_S + (1-y)P_V + (1-x)P_D, \end{aligned} \quad (7)$$

$$\begin{aligned} U_{32} &= xy(R_B - C_B + r) + (1-x)y(R_B - C_B - \theta P_S) + x(1-y)(R_B - C_B - \theta P_S) + (1-x)(1-y)(R_B - C_B - \theta P_S) \\ &= R_B - C_B + (xy - 1)\theta P_S + xyr, \end{aligned} \quad (8)$$

and

$$\bar{U}_3 = zU_{31} + (1-z)U_{32}. \quad (9)$$

Based on replicator dynamics theory [1], we derive the evolutionary dynamics equations for the strategy probabilities of data, validator, and proxy nodes. The growth rates of probabilities x , y , and z for honest strategies among data, validator, and proxy nodes are proportional to the differences between their respective payoffs U_{11} , U_{21} , U_{31} and average payoffs $\bar{U}_1$, $\bar{U}_2$, $\bar{U}_3$. Then the evolutionary dynamics of strategy probabilities for data, validator, and proxy nodes are

TABLE I: Payoff matrix

				VN	
				H	M
PN	S	DN	H	$(\phi R_D - C_D, R_V - C_V, R_B - C_B - C_S)$	$(\phi R_D - C_D, -C_V - P_V, R_B - C_B - C_S + P_V)$
			M	$(-C_D - P_D, R_V - C_V, R_B - C_B - C_S + P_D)$	$(-C_D - C_C - P_D, -C_V - P_V, R_B - C_B - C_S + P_D + P_V)$
	NS	DN	H	$(\phi R_D - C_D + r, R_V - C_V + r, R_B - C_B + r)$	$(\phi R_D - C_D, R_V - C_V, R_B - C_B - \theta P_S)$
			M	$(\phi R_D - C_D, R_V - C_V, R_B - C_B - \theta P_S)$	$(R_{MD} - C_D - C_C, R_{MV} - C_V, R_B - C_B - \theta P_S)$

given by $F_D(x)$, $F_V(y)$, and $F_P(z)$, respectively, expressed as

$$\begin{aligned} F_D(x) &= \frac{dx}{dt} = x(U_{11} - \bar{U}_1) \\ &= -x(x-1)[C_C - R_{MD} - yC_C + \phi R_D + zP_D \\ &\quad + yR_{MD} + zR_{MD} + yr - y\phi R_D - yzR_{MD} \\ &\quad - ryz + yz\phi R_D], \end{aligned} \quad (10)$$

$$\begin{aligned} F_V(y) &= \frac{dy}{dt} = y(U_{21} - \bar{U}_2) \\ &= -y(y-1)[R_V - R_{MV} + xR_{MV} + zP_V - xR_V \\ &\quad + zR_{MV} + xr - xzR_{MV} + xzR_V - xzr], \end{aligned} \quad (11)$$

and

$$\begin{aligned} F_P(z) &= \frac{dz}{dt} = z(U_{31} - \bar{U}_3) \\ &= z(z-1)C_S - P_D - P_V - \theta P_S + xP_D + yP_V \\ &\quad + xyr + xy\theta P_S. \end{aligned} \quad (12)$$

II. THEORETICAL ANALYSIS AND PROOFS

This appendix provides detailed proofs for the theoretical results presented in Section VI.A.

A. BLS-PBFT Communication Complexity Analysis

Theorem II.1 (Communication Complexity Reduction). *The BLS-PBFT consensus mechanism reduces message complexity from $O(n_v^2)$ of traditional PBFT to $O(n_v)$ while maintaining Byzantine fault tolerance.*

Proof. In traditional PBFT with n_v validator nodes:

- Pre-prepare phase: $n_v - 1$ messages
- Prepare phase: $n_v(n_v - 1)$ messages
- Commit phase: $n_v(n_v - 1)$ messages
- Total: $2n_v^2 - n_v - 1 = O(n_v^2)$ messages

In BLS-PBFT (Section IV.E of main text):

- Step E-3: n_v validators send signatures to proxy node: n_v messages
- Step E-4: Signature aggregation $sig_v^{agg} = \prod_{i=1}^{cnt} sig_{v,i}$ (local computation)
- Step E-5: Proxy broadcasts aggregated signature to n_v validators: n_v messages
- Total: $2n_v = O(n_v)$ messages

The key lies in BLS signature aggregation that compresses multiple signatures into a fixed-size signature, with verification requiring only one pairing operation:

$$e(sig_v^{agg}, G) = e\left(H(m), \prod_{i=1}^{cnt} pk_i\right). \quad (13)$$

Thus achieving order-of-magnitude complexity reduction. $\square$

B. Dynamic Threshold Security Analysis

Lemma II.2 (Weighted Threshold Intersection). *Let n consensus nodes have total normalized weight 1, with Byzantine nodes having total weight β_w . If the consensus threshold weight $\alpha_w > (1 + \beta_w)/2$, then any two signature sets with weight at least α_w must contain at least one common honest signer.*

Proof. Let Q_1 and Q_2 be two signature sets with $w(Q_1) \geq \alpha_w$ and $w(Q_2) \geq \alpha_w$. By set theory:

$$w(Q_1 \cup Q_2) = w(Q_1) + w(Q_2) - w(Q_1 \cap Q_2). \quad (14)$$

Since $w(Q_1 \cup Q_2) \leq 1$ (total weight upper bound), we have

$$w(Q_1 \cap Q_2) \geq w(Q_1) + w(Q_2) - 1 \geq 2\alpha_w - 1. \quad (15)$$

When $\alpha_w > (1 + \beta_w)/2$,

$$w(Q_1 \cap Q_2) \geq 2\alpha_w - 1 > 2 \cdot \frac{1 + \beta_w}{2} - 1 = \beta_w. \quad (16)$$

Even if all Byzantine nodes are in the intersection, the remaining weight $(w(Q_1 \cap Q_2) - \beta_w) > 0$ must come from honest nodes. $\square$

Theorem II.3 (Dynamic Threshold Security). *The dynamic threshold κ defined in Eq. (23) of the main text guarantees system security by preventing blockchain forks.*

Proof. *Step 1: Establish security condition.* Assume two different blocks B and B' are committed at the same height h . According to the protocol, each block requires at least κ validator signatures. Let Q_B and $Q_{B'}$ be the signing sets.

Define node weight as normalized reputation score:

$$w_i = \frac{RS_i}{\sum_j RS_j}. \quad (17)$$

The total weight of signing sets satisfies:

$$w(Q_B) \geq \frac{\kappa}{n} \cdot \frac{1}{n} \sum_{i \in Q_B} \frac{RS_i}{NS}. \quad (18)$$

Step 2: Estimate Byzantine weight upper bound. Using Cantelli's inequality to estimate the proportion of low-reputation nodes. For reputation score X with mean $\mu = \overline{NS}$ and variance $\sigma^2 = \sigma_{NS}^2$:

$$P(X \leq \mu - t) \leq \frac{\sigma^2}{\sigma^2 + t^2}. \quad (19)$$

Setting the malicious node reputation threshold as $\overline{NS} - t$, choosing t such that $P(X \leq \overline{NS} - t) \leq \chi$. Solving yields:

$$t \approx \sigma_{NS} \sqrt{\frac{1 - \chi}{\chi}}. \quad (20)$$

Therefore, the expected total Byzantine weight:

$$\beta_w \approx (1 - \chi \overline{NS})^\gamma. \quad (21)$$

where γ reflects the skewness of reputation distribution.

Step 3: Determine dynamic threshold. According to Lemma II.2, security requires:

$$\frac{\kappa}{n} > \frac{1 + \beta_w}{2}. \quad (22)$$

Incorporating reputation variance for robustness:

$$\kappa = \varsigma n(1 - \chi \overline{NS}) \gamma \sigma_{NS}^2. \quad (23)$$

where ς ensures the security condition is satisfied under all network states.

Step 4: Adaptivity analysis. The threshold dynamically adjusts: decreasing in high-reputation scenarios for improved efficiency while increasing in low-reputation scenarios for enhanced security, achieving optimal trade-off between efficiency and security. $\square$

C. Reputation System Convergence Analysis

Theorem II.4 (Reputation Score Convergence). *The reputation update mechanism in Eq. (30) of the main text converges to stable values accurately reflecting node behavior within $O(T_u \cdot \log(1/\epsilon))$ update periods.*

Proof. Construct the Lyapunov function:

$$V(t) = \sum_{v \in N} [RS(v, t) - RS^*(v)]^2. \quad (24)$$

where $RS^*(v)$ is the true reputation of node v .

For honest nodes ($P(v, t) = 0$),

$$\Delta V_h = -2K \cdot S(v, t) \cdot f(C_h) \cdot [RS(v, t) - RS^*(v)] < 0. \quad (25)$$

For malicious nodes ($P(v, t) > 0$),

$$\Delta V_m = -2 \exp(P(v, t)) \cdot [RS(v, t) - RS^*(v)] < 0. \quad (26)$$

Thus $V(t)$ decreases monotonically, converging at exponential rate:

$$V(t) \leq V(0) \cdot \exp(-\lambda_r t / T_u). \quad (27)$$

where $\lambda_r = \min\{2K, 2\}$. For ϵ -convergence: $t = O(T_u \cdot \log(V(0)/\epsilon))$. $\square$

D. Evolutionary Game Stability Analysis

Theorem II.5 (Global Stability). *When system gain $r > \max\{R_{MD} - \phi R_D - C_C, R_{MV} - R_V, C_S\}$, the evolutionary game system globally converges to the ideal equilibrium $E_7(1, 1, 0)$.*

Proof. *Step 1: Local stability analysis.* At equilibrium point $E_7(1, 1, 0)$, the Jacobian matrix eigenvalues are:

$$\lambda'_1 = -r, \quad \lambda'_2 = -r, \quad \lambda'_3 = -(C_s + r). \quad (28)$$

When $r > 0$ and $C_s > 0$, all eigenvalues are negative, confirming local asymptotic stability.

Step 2: Global stability proof. Construct relative entropy as Lyapunov function:

$$H(x, y, z) = x \ln \frac{x}{1} + y \ln \frac{y}{1} + (1 - z) \ln \frac{1-z}{1}. \quad (29)$$

Step 3: Prove $\dot{H} < 0$. Under the stated condition:

- When $x < 1$: $F_D(x) > 0 \Rightarrow (x - 1)F_D(x) < 0$
- When $y < 1$: $F_V(y) > 0 \Rightarrow (y - 1)F_V(y) < 0$
- When $z > 0$: $F_P(z) < 0 \Rightarrow -zF_P(z) < 0$

For all $(x, y, z) \neq (1, 1, 0)$, we have $\dot{H} < 0$.

Step 4: Apply LaSalle's invariance principle. The system converges to the largest invariant set where $\dot{H} = 0$, which is $E_7(1, 1, 0)$. The convergence rate is exponential with time constant $\tau = 1 / \min\{r, C_s\}$. $\square$

REFERENCES

- [1] T. Mai, H. Yao, N. Zhang, L. Xu, M. Guizani, and S. Guo, "Cloud mining pool aided blockchain-enabled Internet of Things: An evolutionary game approach," *IEEE Trans. Cloud Comput.*, vol. 11, no. 1, pp. 692–703, Jan.-Mar. 2023.